The stator is made by the insulated silicon steel plate shown in figure 4.12. Both the primary and the secondary winding are fixed with a $90^0$ electric angle.

## ii. Rotor

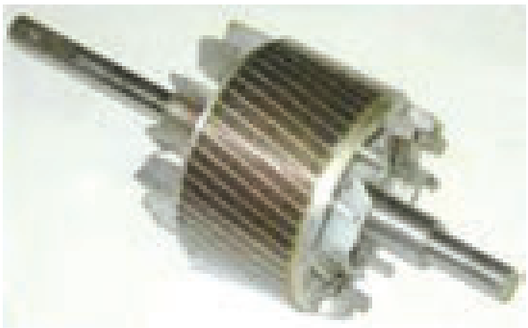


Fig. 4.13 **Rotor**

The structure of a squirrel cage rotor is as shown in figure 4.13. This type of rotor is fixed inside the stator.

## iii. Blades



Fig. 4.14 **blades**

The blades are made up of steel or aluminum sheet, as shown in figure 4.14. Usually three or four blades are fitted in this type of fan. The blades are fixed on the shaft with the help of screws.

## iv. Sleeve bearing

Sleeve bearings are mounted on shaft of the fan. These bearings (as in figure 4.15) are used to reduce friction in rotating parts and reduce the sound while in rotation.



Fig. 4.15 **Sleeve bearing**

## v. Oscillating mechanism

The oscillating mechanism is connected to the back of the electric motor shaft. The wrenching system is used to turn the left and right back into the side of the waist to a certain angle.

## b) Speed regulator

The speed regulator used is of resistant type which is connected in series with the fan and required flow of air can be utilised.

## c) Base of the fan

The bottom of the fan is made of steel iron. When an electric motor rotates, the base is fixed and stay in the same place, even the fan is in tilting movement.

## d) Working principle

When an electric supply is given to the fan, the current passes to the main winding and secondary winding and produces rotating magnetic field. The 2.5 micro farad capacitor is connected in series with the secondary winding as in figure 4.16. Due to the production of rotating magnetic field, the blades connected with the rotor rotate and air flow will be circulated to the area and direction where it is fixed. Usually the



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